

TIMOTHY POORMAN

Mechanical Engineer (B.S. May 2026)

Mechanical Engineer with hands-on experience in mechanical design, engineering testing, instrumentation, data analysis, and manufacturing. Experienced with SolidWorks, LabVIEW, NI DAQs, root-cause analysis, and engineering calculations involving thermodynamics, heat transfer, and fluid mechanics. Strong background in developing data-driven solutions to engineering and product problems through testing, analysis, design iteration, and collaboration.

Technical Skills

- Mechanical Design: SolidWorks, Machine Design, Stress Calculations, Design Iteration
- Engineering Analysis: Thermodynamics, Fluid Mechanics, Data Analysis, Root-Cause Analysis
- Testing & Instrumentation: NI DAQs, RPT/RTD Sensors, Motor Testing, Motor Alignment, LabVIEW
- Fabrication & Prototyping: 3D Printing, Mechanical Fabrication, Fluids Lab Modules
- Software & Programming: Excel, Python, C++, MatLab, Arduino

Work Experience

Scot Industries-Sales Engineer | June 2026 – September 2026

- Collaborated with plant supervisor on manufacturing process improvements to streamline production methods and improve efficiency.
- Performed root-cause analysis on customer product issues to identify manufacturing and design-related failures. Implemented an inventory system to reduce material waste within the manufacturing process.
- Designed a lifting device in SolidWorks and performed engineering stress calculations to evaluate the design under specified loading conditions for 2600 pounds.
- Developed machine timers to track operator efficiency and supported additional engineering projects.

Komatsu Mining-Engineering Intern | August 2024-April 2026

- Conducted SR motor testing and optimization using NI DAQs, RPT/RTD sensors, and motor alignment tools.
- Collected and interpreted engineering test data to evaluate motor performance and reliability; supported a facility performing several hundred motor tests annually.
- Used LabVIEW, proprietary software, and Excel for data acquisition, analysis, and interpretation.
- Supported engineering projects involving SolidWorks, Excel, thermodynamics, and heat transfer.

ENGINEERING PROJECTS

Instrumentation & Data Acquisition

- Worked with NI DAQs, RPT/RTD sensors, and motor alignment tools for engineering data acquisition and analysis using LabVIEW, proprietary software, and Excel.

Heat Transfer & Design Optimization — Weld-On Sensitive Component

- Supported root-cause analysis of overheating in temperature-sensitive welds.
- Applied heat-transfer and thermodynamic principles to approximate heat penetration and develop design/process recommendations.

Process Optimization — Joyal Lug Fuser

- Collected and analyzed test data to optimize current and compression settings, improving consistency and quality of fused lugs.

Fluid Systems Engineering — Headloss System

- Led an engineering class sub-team in researching, designing, and building a pressure headloss system. Applied fluid-mechanics principles and engineering calculations throughout the design process.

SolidWorks Design & Fabrication

- Designed modular storage systems for sensitive electronics, controller mounts, and frame supports.
- Designed and fabricated a pen plotter using 3D-printed components, Arduino hardware/programming, and a servo-motor pulley system.

Education

LeTourneau University-Longview, TX

B.S. Mechanical Engineering (May 2026)

Minors: Mathematics and Applied Science

Relevant Coursework: Mechanics of Materials, Thermodynamics, Heat Transfer, Fluid Mechanics, Manufacturing Processes, Machine Design

Location

LONGVIEW, TX

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Portfolio

